Microbiome case study

Full pooled data | Original preprocessing | D, DT, ICFM, MBGAN

Two-sample Comparison through Additive Tree Models for Density Ratios

1,458
real observations

123
retained taxa

1,000
samples / generator

Final analysis

The original pooled real matrices contain **1,166 training and 292 test observations**. The original preprocessing, taxon order and observation split are preserved. All four generators are fitted from these training rows using released code and frozen recipes, presented in the order **D, DT, ICFM, MBGAN**. No real test rows enter generator fitting or selection.

Eight real-versus-generated comparisons and the held-out null experiment use BATTS **0.3.0**, commit **147e511d**. Gradient and Hellinger boosting explicitly use all fitting observations at each iteration (subsample_fraction = 1). Figures 6, 7 and S8-S12 summarize this run.

Real-test pointwise posterior summaries

Generator	95% intervals excluding zero	Mean absolute posterior-mean log ratio
D	210/292 (71.9%)	1.235
DT	174/292 (59.6%)	0.605
ICFM	291/292 (99.7%)	1.699
MBGAN	292/292 (100.0%)	5.676

These are descriptive pointwise posterior summaries, not global two-sample p-values. Intervals containing zero do not establish distributional equality. Each density-ratio comparison is fitted separately.

Report contents

Page 2: methods and generator provenance.
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Page 10: data sources, verification and reproducibility.

Methods and generator provenance

Preserved real inputs; all generators fitted from the training rows

Generator constructions

Generator	Construction
D	Dirichlet Newton maximum-likelihood fit on the 1,166 training rows; seed 1.
DT	Positive-parameter Beta maximum-likelihood fits at 122 tree nodes, using the 1,166 training rows; seed 1.
ICFM	CPU; random initialization; MLP with three hidden layers of width 64; 20,000 Adam updates, batch 256, rate 0.001, flow noise 0.0001; training seed 20261003; final update.
MBGAN	GPU; random initialization; 500,000 generator iterations, five critic updates each, batch 32; RMSprop rate 0.00005; training seed 1; final iteration. Gradient penalty is differentiated through critic weights.

All four fits use only the preserved 1,166 training rows. Neural training starts from seeded random initialization; no historical weights are required. Training length and sampling settings are frozen before evaluation; the final update is used without test-based selection. Two full independent runs must agree within the recorded hardware and software environment; equality across CPU and GPU is not assumed.

BATTS settings

Seed 1 per estimator; 200 Bayesian trees; 2,000 burn-in and 1,000 retained iterations; thinning 1; fixed lambda 5; margin 0.1. Boosting uses 5-fold CV, up to 1,000 trees, depth 4, learning rate 0.01, 32 bins, and subsample_fraction = 1. The figures show Bayesian results.

Training comparison: intervals excluding zero

Generator	Real training rows	Generated rows
D	1166/1166	1000/1000
DT	1153/1166	966/1000
ICFM	1162/1166	964/1000
MBGAN	1166/1166	1000/1000

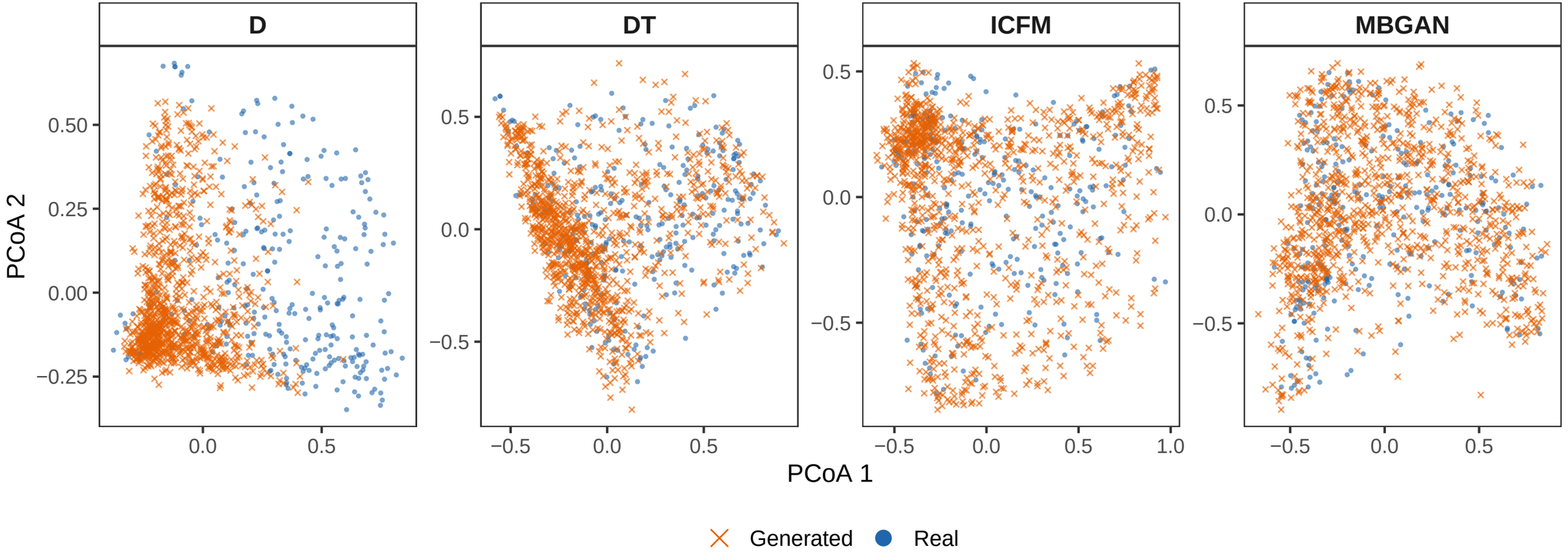
Interpretation

The test comparisons fit density ratios using the real test rows themselves; they are not predictions from training-comparison estimators. Different reference sizes limit train/test comparisons. Filtering precedes the observation-level split; repeated observations may share participants across splits; participant-level separation is not established.

8/16 boosting fits selected the 1,000-tree search limit. This follows the supplied single-chain workflow; no new multi-chain convergence assessment is claimed.

Figure 6 | Test-data ordination

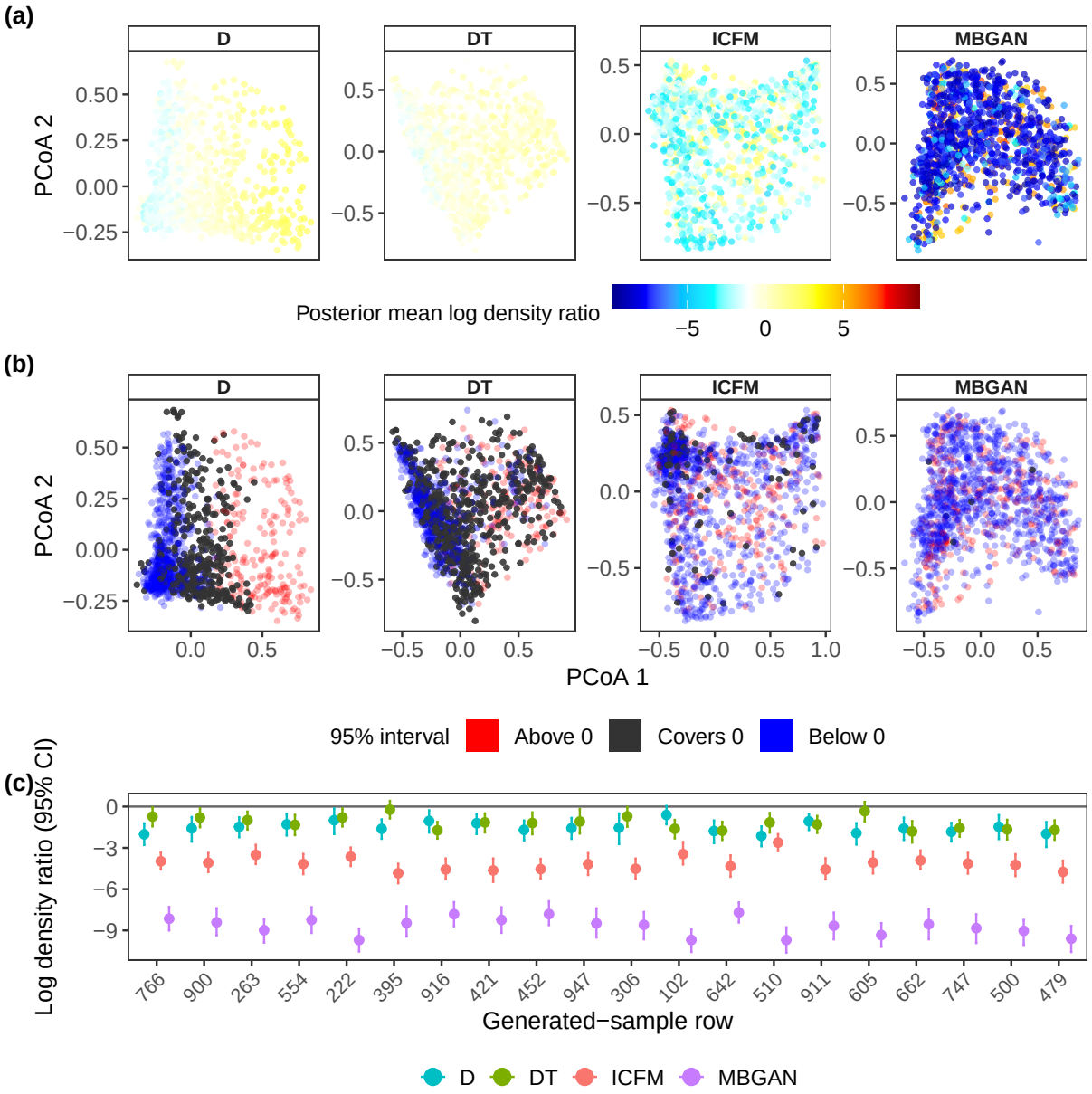
Original pooled data | BATTS 147e511d



Real test compositions versus four generators. Each panel contains 292 real observations (blue dots) and 1,000 generated observations (orange crosses). The first two principal coordinates use Bray-Curtis dissimilarities with an additive correction. Each panel is a separate joint ordination of real and generated data; axes are not shared across generator panels. Visual overlap in two dimensions does not establish equality of the full compositional distributions.

Figure 7 | Test-data density ratios

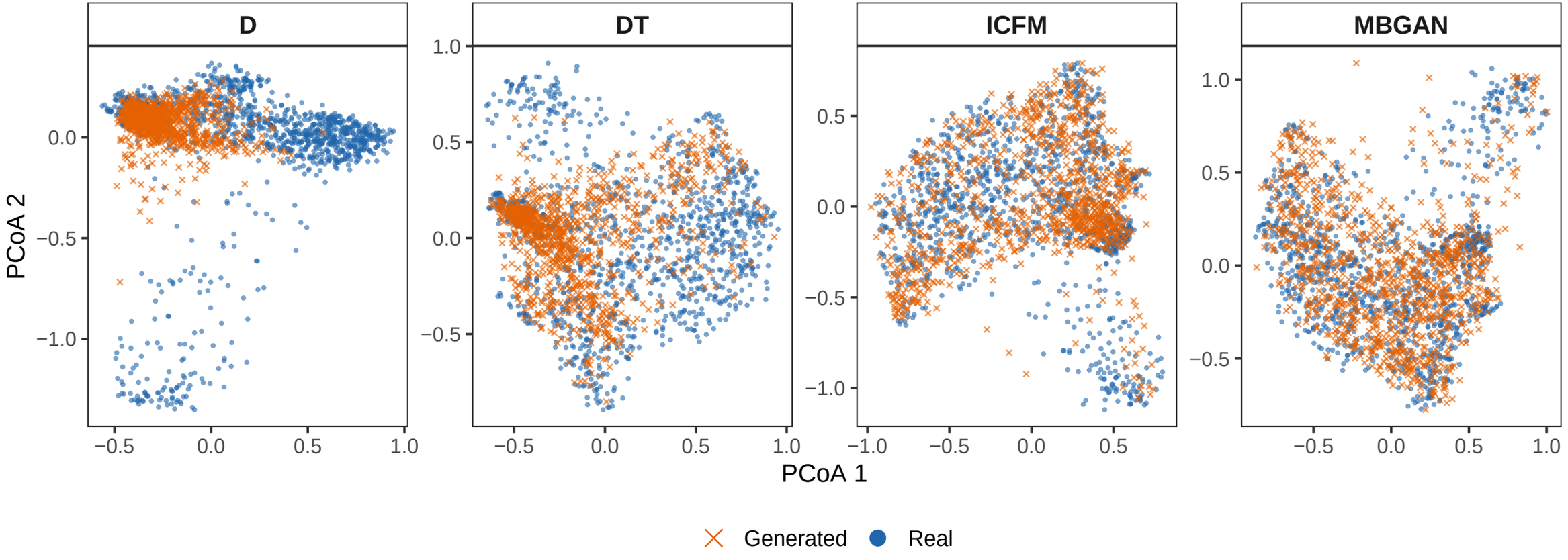
Original pooled data | BATTS 147e511d



Bayesian density-ratio summaries for the test comparisons. (a) Posterior mean log density ratio on the corrected Bray-Curtis ordination. (b) Classification of central 95% intervals as above zero, covering zero, or below zero. (c) Posterior means and intervals at 20 generated-row indices selected by the average absolute posterior mean across the four generator comparisons. These are selected extremes. Display labels are row positions within the generated sample banks. The same index across generators labels different generated compositions, not the same observation evaluated under four generators.

Figure S8 | Training-data ordination

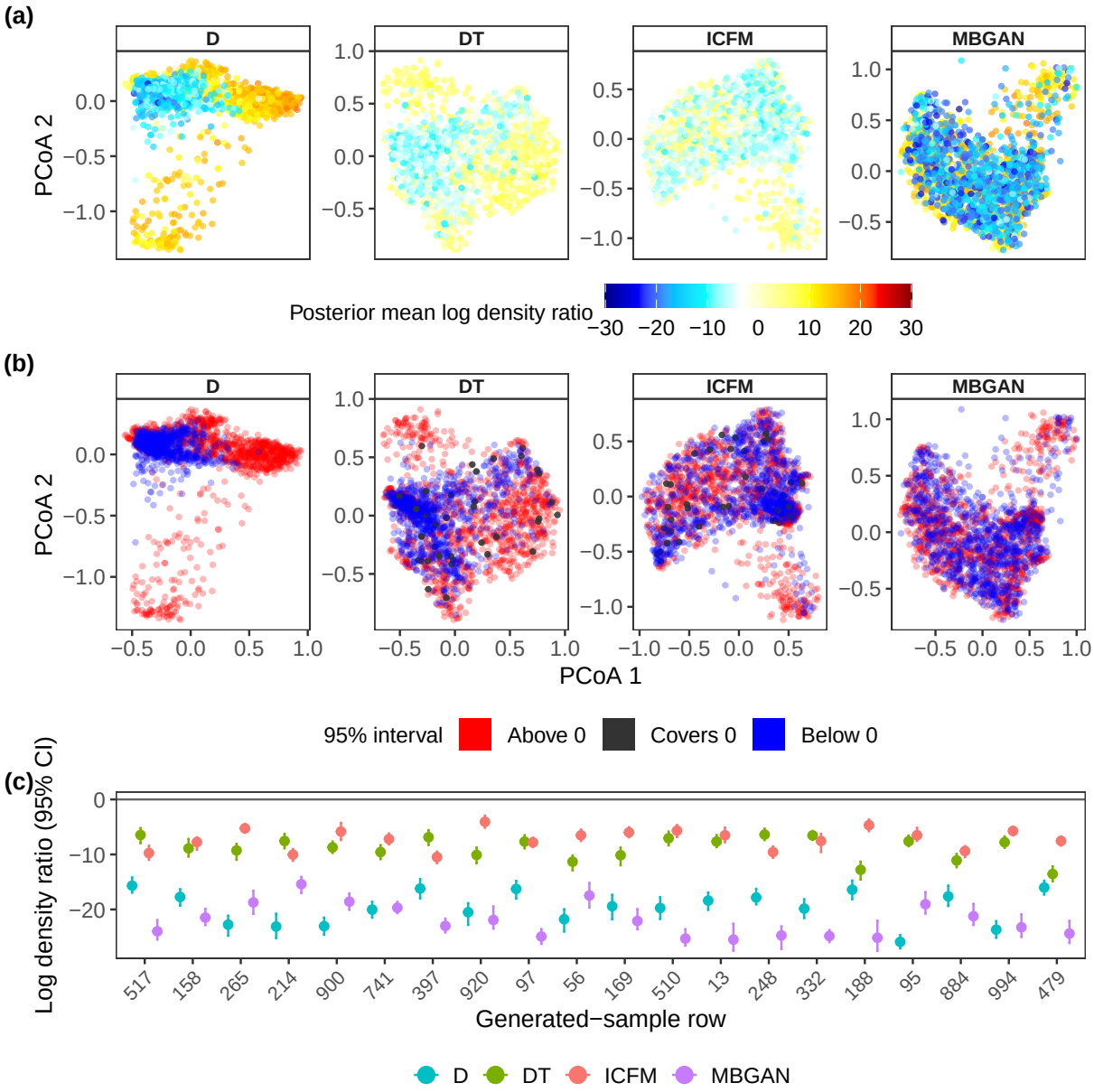
Original pooled data | BATTS 147e511d



Real training compositions versus four generators. Each panel contains 1,166 real observations (blue dots) and 1,000 generated observations (orange crosses). The first two principal coordinates use Bray-Curtis dissimilarities with an additive correction. Each panel is a separate joint ordination of real and generated data; axes are not shared across generator panels. Visual overlap in two dimensions does not establish equality of the full compositional distributions.

Figure S9 | Training-data density ratios

Original pooled data | BATTS 147e511d



Bayesian density-ratio summaries for the training comparisons. (a) Posterior mean log density ratio on the corrected Bray-Curtis ordination. (b) Classification of central 95% intervals as above zero, covering zero, or below zero. (c) Posterior means and intervals at 20 generated-row indices selected by the average absolute posterior mean across the four generator comparisons. These are selected extremes. Display labels are row positions within the generated sample banks. The same index across generators labels different generated compositions, not the same observation evaluated under four generators.

Figure S10 | Training posterior mean and quantiles

Original pooled data | BATTS 147e511d

Posterior summaries across the training compositions. Columns correspond to D, DT, ICFM and MBGAN, in that order. Rows show the posterior mean, lower 2.5% quantile and upper 97.5% quantile of the log density ratio. Real samples are dots; generated samples are crosses. These grids use classical Bray-Curtis PCoA without an additive correction, following the legacy workflow, and a symmetric-log color scale. Their coordinates differ from those in Figures 6, 7, S8 and S9. The plotted color limits are -30 to 30.

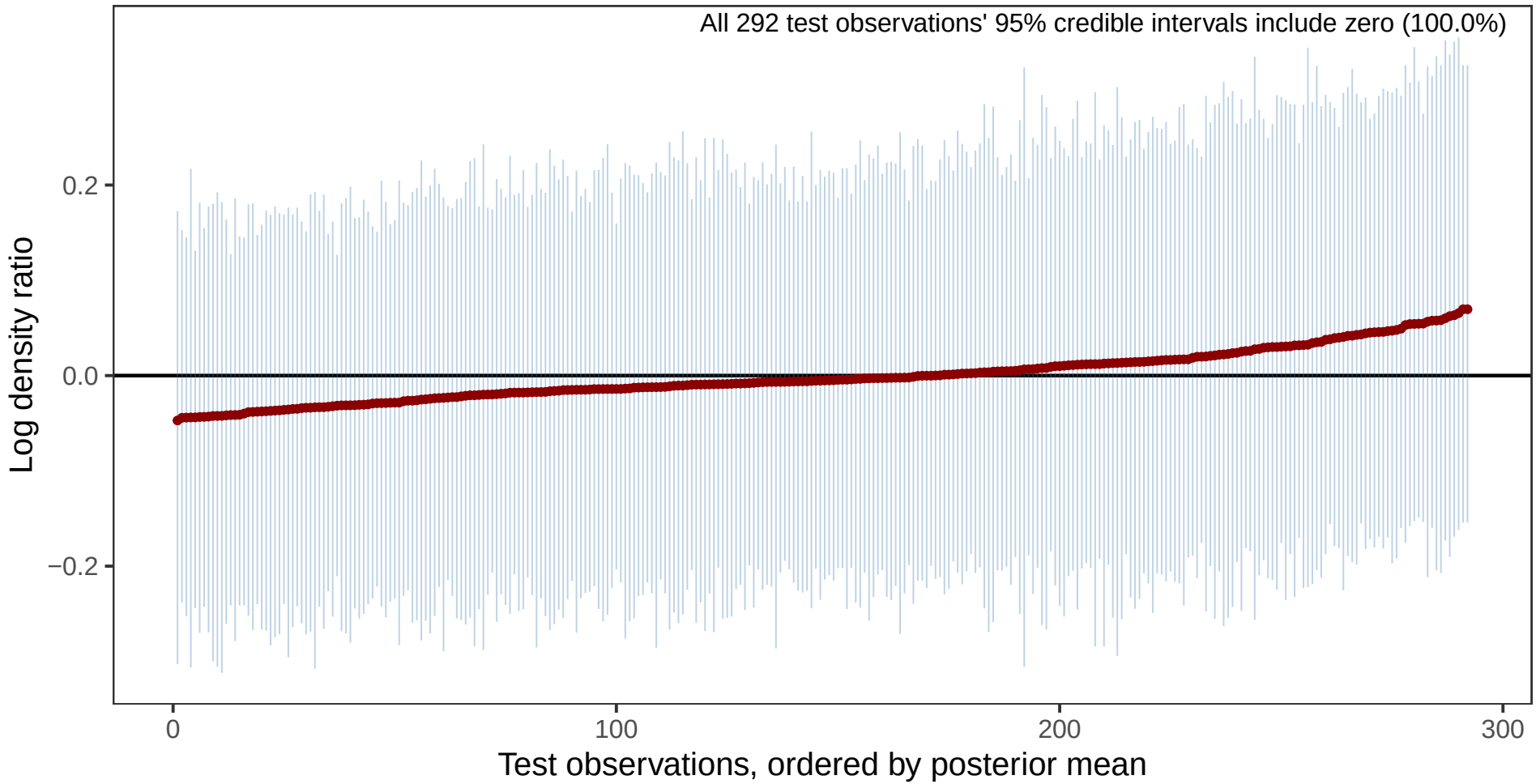
Figure S11 | Test posterior mean and quantiles

Original pooled data | BATTS 147e511d

Posterior summaries across the test compositions. Columns correspond to D, DT, ICFM and MBGAN, in that order. Rows show the posterior mean, lower 2.5% quantile and upper 97.5% quantile of the log density ratio. Real samples are dots; generated samples are crosses. These grids use classical Bray-Curtis PCoA without an additive correction, following the legacy workflow, and a symmetric-log color scale. Their coordinates differ from those in Figures 6, 7, S8 and S9. The plotted color limits are -12 to 12.

Figure S12 | Held-out null diagnostic

Original pooled data | BATTS 147e511d



Real-versus-real null experiment. Balanced random labels are assigned to the 1,166 real training observations (seed 2026). The Bayesian fit uses 200 trees, 500 burn-in and 500 retained iterations, fixed lambda 5 and margin -1. The 292 held-out real test observations are ordered by posterior mean. 292/292 pointwise 95% credible intervals contain zero. This is the observed result of one null experiment, not a guarantee of nominal coverage.

Data sources and reproducibility

Original preprocessing preserved through frozen real matrices

Real-data composition

Study	Training	Test	Total
HMP_2019_ibdmdb	751	180	931
HMP_2019_t2d	210	67	277
HallAB_2017	171	38	209
HanniganGD_2017	34	7	41

The original curatedMetagenomicData selection is normalized over 123 retained taxa and stored at float32 precision. Counts refer to observations, not unique participants. The recovered sample-selection manifest, taxon order and preprocessing code accompany the release. Filtering is collection-wide, before the observation split. Real input hashes are verified.

Execution and verification

All six input matrices are checked for dimensions, finite nonnegative values and row sums. All eight estimator tables are checked for dimensions, finite values, ordered posterior quantiles and agreement with numerical summaries. Seven source figures are embedded as PDF content. Every report page is rendered and checked for decoding warnings and text outside page bounds before COMPLETE is written. Generator training, sample hashes and reproduction evidence must also pass the provenance gate.

The separate null fit has **292/292** pointwise 95% intervals containing zero. Balanced random real-training labels use seed 2026; the fit uses 200 trees, 500 burn-in and 500 retained iterations, fixed lambda 5 and margin -1, evaluated on the real test matrix. One null experiment does not establish a repeated-sampling coverage guarantee.

Portable run artifacts

Input and source hashes: input_manifest.json. Generator evidence: generation/. Fitting settings: settings.json and settings.R. Numerical results: interval_summary.csv and boosting_cv_summary.csv. Estimator tables: revision/. Figures: figures/. Verification: analysis_validation.json and report_manifest.json.

BATTS 0.3.0: `147e511d211919afefb8decac6333e66d24919db`
Workflow: github.com/yuliangxu/TwoSample. Workflow source snapshots and SHA-256 records are in the input manifest.

References for the real data

Pasolli et al. (2017). Accessible, curated metagenomic data through ExperimentHub. Nature Methods 14:1023-1024. doi:10.1038/nmeth.4468

Lloyd-Price et al. (2019). Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases. Nature 569:655-662. doi:10.1038/s41586-019-1237-9

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Hall et al. (2017). A novel Ruminococcus gnavus clade enriched in inflammatory bowel disease patients. Genome Medicine 9:103. doi:10.1186/s13073-017-0490-5

Hannigan et al. (2018). Diagnostic Potential and Interactive Dynamics of the Colorectal Cancer Virome. mBio 9:e02248-18. doi:10.1128/mBio.02248-18